

Rayna Yu

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EDUCATION

Northeastern University

Boston, MA

B.S in Computer Science and Mathematics

May 2028

GPA: 4.0/4.0 | Honors: Dean's List, Dean's Scholarship

Relevant Coursework: Object-oriented design, Algorithms (Graduate-level), Artificial Intelligence, Theory of Computation, Matrix Methods in Data Analysis and Machine Learning, Linear Algebra, Probability and Statistics

Activities: Amazon IET-MSI program participant, Women in Tech, NUAerospace, Forge Product Lab, Oasis

SKILLS

Languages: Java, Python, Matlab, JavaScript, TypeScript, SQL, HTML/CSS, Racket, OCaml, C++

Frameworks: React, Next.js, FastAPI, PgAdmin 4, NumPy, Scikit-learn, Pandas, Matplotlib, Turf.js, TensorFlow

Tools & Technologies: Git/GitHub, Visual Studio Code, IntelliJ, Eclipse IDE, Linux

EXPERIENCE

Code 4 Community

January 2025 - Present

Software Developer

Boston, MA

- Developed full-stack applications for nonprofits (GI Boston, Securing Safe Food, NEFAC, etc.), converting requirements into functional sites using **React**, **Next.js**, and **PostgreSQL**, improving data accessibility and operational efficiency
- Collaborated in a 7-person Agile team to streamline backend operations for a recruitment portal, implementing database optimizations with **pgAdmin 4**, and contributing through sprint planning, **Git/GitHub** workflows, and peer code reviews

Teuscher Lab

May 2025 - August 2025

Software Engineer and Researcher (paper available upon request)

Remote

- Developed a **React Native** mobile navigation app in **TypeScript**, integrating multi-source **GeoJSON** datasets, **Turf.js**, **OpenRouteService API**, and **FastAPI** backend to optimize routes by detecting pedestrian-specific hazards
- Implemented and evaluated multiple machine learning models to predict vehicle-pedestrian crashes using **Python**, **Scikit-learn**, **Pandas**, **NumPy**, and **Matplotlib**, selecting **Random Forest** as the top-performing model with 94% accuracy

Northeastern University; Computer Science Department

May 2025 - Present

Teaching assistant, CS 2800: Logic and Computation and CS1800: Discrete Structures

Boston, MA

- Supported 100+ students in CS 2800 & CS 1800, communicating complex concepts effectively while covering property-based testing, code specification, set theory, data structures, and discrete mathematics using **Python** and **OCaml**
- Led weekly recitations and office hours, mentoring students in formal reasoning, proofs, and computational logic

RISD Museum

September 2022 - June 2024

Museum Curator Intern

Providence, RI

- Partnered with a team of 12 peers to design and author the first fully teen-led exhibit, [Listen!](#)
- Coordinated with museum staff and interviewed artists to translate complex societal themes into engaging and accessible experiences, drawing over 5,000 visitors in its first week

PROJECT EXPERIENCE

Horizon Lab/NUAerospace - C++, Matlab, Simulink, Embedded Systems

August 2025 - Present

Flight Software Developer

- Contributed to hands-on engineering development projects for Arduino-based flight software in C++ to monitor rocket internal states in real time, ensuring reliable operation and enabling precise fault detection and localization during launches

Robot Path Planner - Matlab, Debugging

July 2025 - Present

Software Engineer

- Designed a 2D robot path planner combining Dijkstra and reinforcement learning for collision-free navigation

Calendar Application - Java, GUI, Swing, Unit testing

May 2025 - June 2025

Java Developer

- Architected a modular calendar application in Java using OOP and OOD principles within an MVC framework

VOLUNTEER EXPERIENCE

Animal Rescue League of Boston

August 2023 – Present

Shelter Support

Boston, MA

- Assist with animal care for 50+ cats and contribute to team efforts to improve animal welfare by evaluating animal behavior